

VENUSPRO — Form Builder Configuration Guide

How to build a metadata-driven insurance **form** end-to-end — from basic details, sections and fields, through conditional logic, formulas, the rule/rating engine, validation and platform guardrails, to live preview, publish and versioning — with field-by-field explanations and screenshots of every flow.

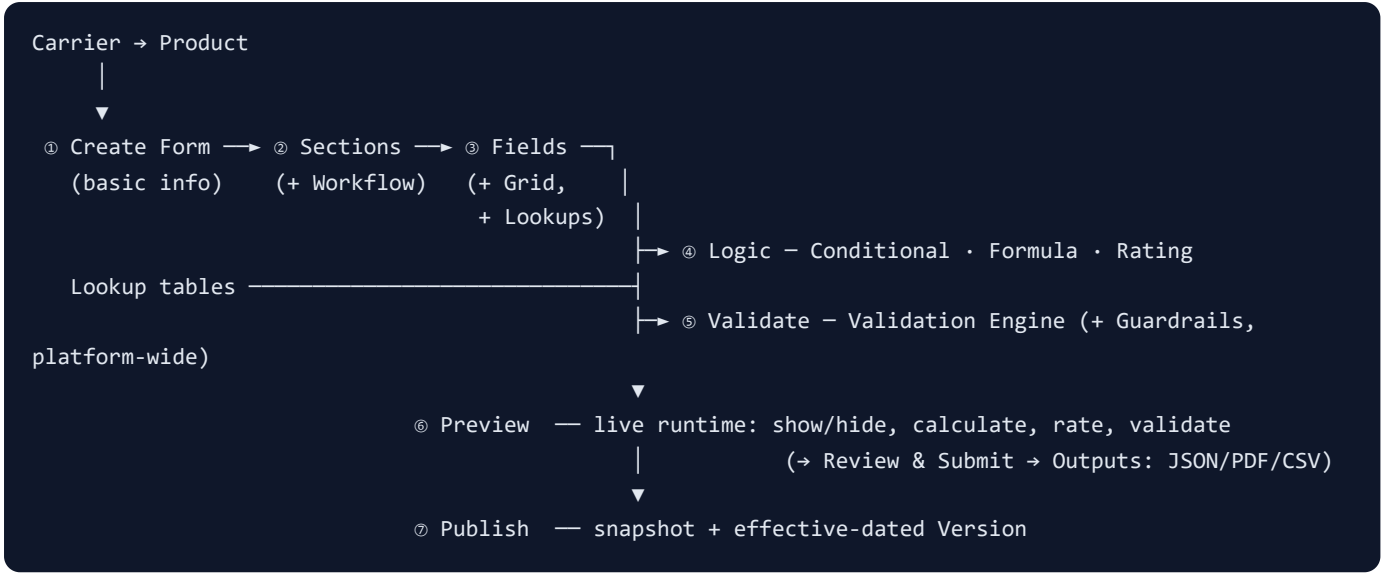
Audience: product/underwriting admins and form designers configuring the questionnaires a carrier uses across the policy lifecycle (Application, Quote, Endorsement, Cancellation, Claim, Renewal...). **Where:** the **Form Designer** and **Catalog** sections of the VENUSPRO admin (sidebar).

1. The mental model — building blocks

Everything in the platform is **metadata**. You don't code a form — you describe it, and the runtime engines render, calculate, rate and validate it live from that description. Get these blocks straight and the rest is easy.

Block	What it is	Analogy	Where you edit it
Carrier / Product	The catalog a form is <i>written for</i> . A Product carries the base rate, coverages and eligible states; a Carrier owns Products.	The catalog	Carriers, Products
Form	The top-level container + its metadata (name, product, type, version, status, dates, navigation style, base premium).	A questionnaire / application	Create Form
Section (+ Subsection)	A group of fields with a layout; the unit of navigation (a wizard step / tab / accordion panel).	A page / step	Section Builder
Field	One captured data point, with a type and properties (label, API name, required, options...).	A question	Field Builder
Rule (<i>Conditional Logic</i>)	IF (conditions) THEN show / hide / require / ... a field or section.	Smart branching	Conditional Logic
Formula	An expression that derives a value live (for computed / premium fields).	A spreadsheet cell	Formula Builder (and a field's <i>Formula</i> property)
Rating Rule (<i>Rule / Rating engine</i>)	IF (conditions) THEN premium / eligibility / tier / referral / message.	A pricing rulebook	Rule / Rating
Validation	A field / section / form / cross-field check, classified blocker / warning / info .	A guard / assertion	Validation Engine
Lookup Table	Reference data (states, class codes, vehicle makes, limits) that feeds dropdowns, rating factors and validations.	A reference sheet	Lookup Tables
Guardrail	A platform-wide underwriting / rating / validation rule enforced on <i>every</i> quote & bind, on top of each form's own logic.	House rules	Guardrails
Version	A status- and effective-dated snapshot of the whole form definition.	A git tag	Versioning
Submission	The runtime artifact — the captured answers + the rated result + the outputs (JSON / PDF / CSV).	The filled-in application	Preview → Review → Outputs

The pipeline (the canonical 7-step build flow shown as a stepper across the top of every builder screen):



Auxiliary builders fold onto the step they belong to: **Workflow & Nav** → *Sections*; **Repeating Grid** and **Lookup Tables** → *Fields*; **Formula Builder** and **Rule / Rating** → *Logic*; **Review** → *Preview*; **Versioning** and **Outputs** → *Publish*.

2. The Form Designer workspace

The sidebar groups the whole surface. You'll spend most of your time in **Form Designer**, dipping into **Catalog** to pick the carrier/product and **Runtime** to test.

Group	Screens
Workspace	Dashboard · Form Library
Catalog	Carriers · Products · Guardrails
Form Designer	AI Generate · Create Form · Section Builder · Field Builder · Conditional Logic · Formula Builder · Validation Engine · Repeating Grid · Lookup Tables · Rule / Rating · Workflow & Nav
Runtime	Preview (Live) · Review & Submit · Outputs
Platform	Versioning · Audit Log · API Layer · Users & Roles · Settings

VENUSPRO
FORM PLATFORM

WORKSPACE

Dashboard

Form Library

CATALOG

Carriers

Products

Guardrails

FORM DESIGNER

AI Generate

Create Form

Section Builder

Field Builder

Conditional Logic

Formula Builder

Validation Engine

Repeating Grid

Lookup Tables

Rule / Rating

Workflow & Nav

RUNTIME

Preview (Live)

Review & Submit

Outputs

DOCUMENTS

Templates

Clause Library

v0.1 - Metadata-Driven

Workspace > Form Library

Q Search forms...

Viewing as Admin

Admin

FORM DESIGNER

Form Library

Create and manage every form across products, lines and versions.

+ Create Form

All 10

Published 6

Draft 3

Archived 1

All Types

Q Search forms...

FORM NAME	TYPE	PRODUCT	VERSION	STATUS	CREATED BY	UPDATED	
Commercial Auto Application frm-commercial-auto	Application	Commercial Auto	v1.1	Published	Admin	2024-05-12	Active Preview ...
Truck Supplemental frm-truck-supp	Supplemental	Commercial Auto	v1.0	Draft	John D.	2024-05-10	Preview ...
Workers Comp Quote frm-wc-quote	Quote	Workers Comp	v2.0	Published	Admin	2024-05-08	Preview ...
Endorsement — Waiver frm-endorsement	Endorsement	General Liability	v1.0	Archived	Sara K.	2024-05-01	Preview ...
General Liability Application frm-gl-app	Application	General Liability	v3.2	Published	Admin	2024-04-22	Preview ...
BOP Renewal Questionnaire frm-bop-renewal	Renewal	Business Owners	v1.4	Draft	Priya N.	2024-04-18	Preview ...
Auto Claim — First Notice of Loss frm-auto-fnol	Claim	Commercial Auto	v2.1	Published	Sara K.	2024-04-09	Preview ...
Mid-Term Endorsement — Add/Remove Vehicle frm-auto-endorse	Endorsement	Commercial Auto	v1.2	Published	John D.	2024-05-15	Preview ...
Policy Cancellation Request frm-auto-cancel	Cancellation	Commercial Auto	v1.0	Published	Sara K.	2024-05-16	Preview ...
Cancellation — Non-Payment Notice frm-gl-cancel	Cancellation	General Liability	v1.1	Draft	Priya N.	2024-05-14	Preview ...

Click any form to load its own data into Preview & the Designer — each type (Application, Cancellation, Endorsement, Claim, Renewal...) opens its own sections and fields. Currently active: Commercial Auto Application.

The **Form Library** lists every form with its **Type**, **Product**, **Version**, **Status** (Draft / Published / Archived) and last-updated date. The tabs (All / Published / Draft / Archived), the **Type** filter and the search box narrow the list. **Click any row's Preview** to load that form's own sections & fields into the Designer and Runtime — each type (Application, Cancellation, Endorsement, Claim, Renewal...) opens its own template. **Create Form** (top-right) starts a fresh one.

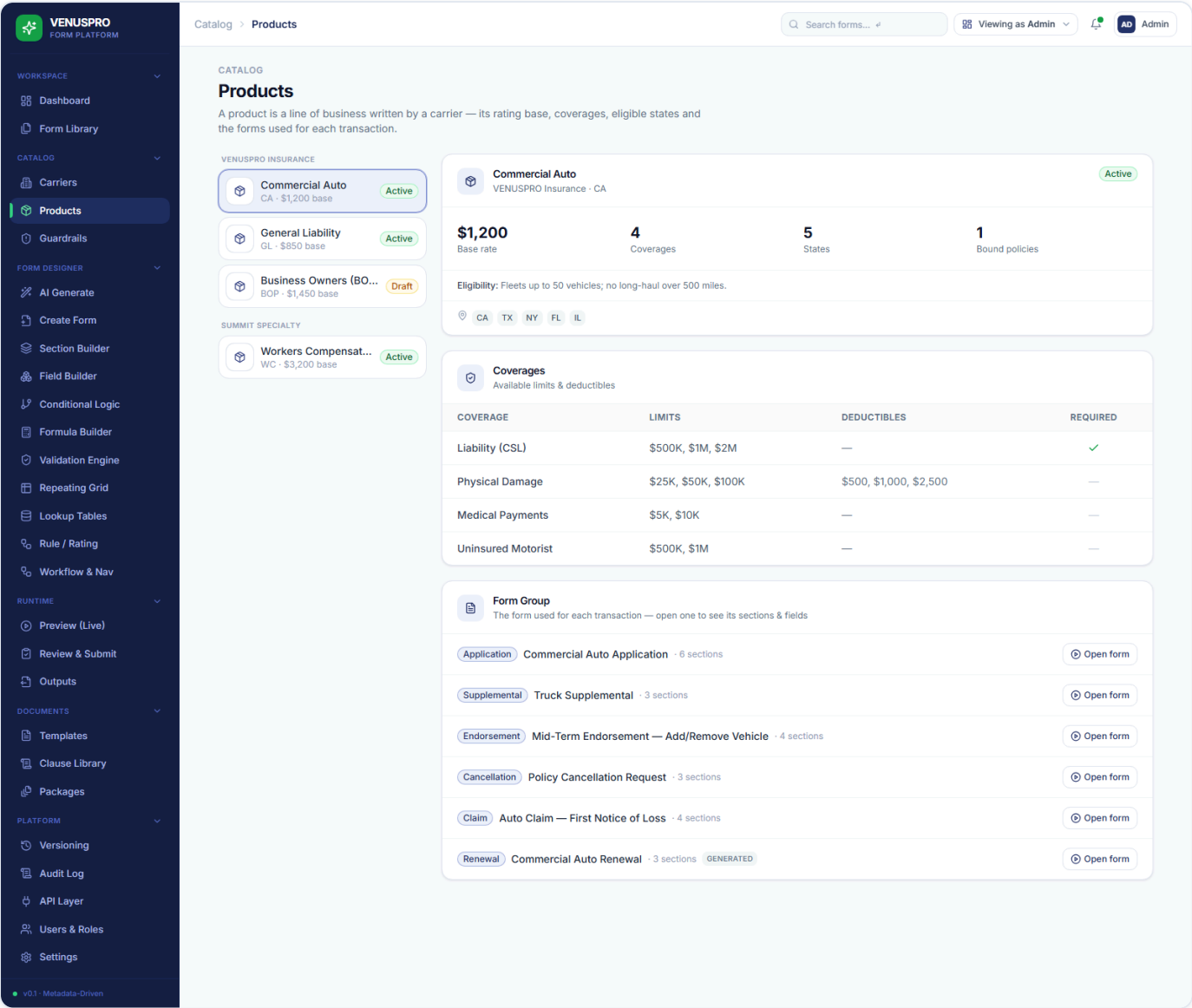
⚠ **One active form at a time.** The Designer holds a single editable form (the "active" form). Loading another form from the Library — or generating one with AI — **replaces** what's currently in the builders. Snapshot first with **Versioning** → **Clone** if you want to keep it.

3. Configuration — step by step

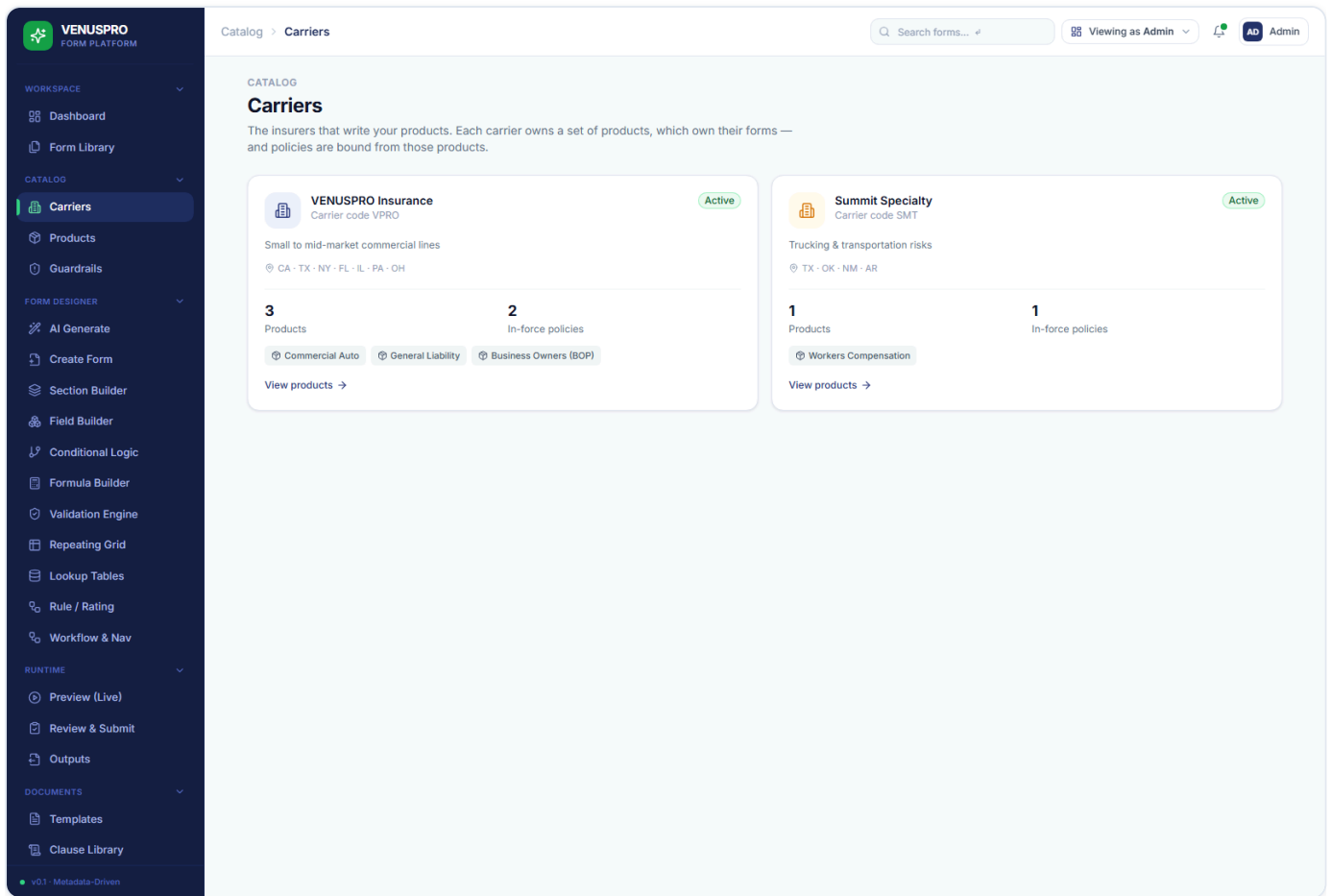
Configure in flow order: **Catalog** → **Create** → **Sections** → **Fields** → **Logic** → **Validate** → **Preview** → **Publish**.

Step 0 — Pick the Catalog (Carrier & Product)

A form is always *written for* a product. The product supplies the base rate, the coverage menu and the eligible states the rest of the form (and the guardrails) lean on.



Products is a master–detail screen. Pick a product on the left to see its **Summary** (base rate, coverage count, states, bound policies), its **Coverages** table (limits, deductibles, required) and its **Form Group** — the form used for each transaction type, each with an **Open form** button that loads it into Preview.



Carriers shows each insurer's appetite, filed states, product count and in-force policies. (Carriers and Products are read-only catalogs today — see *Gaps*.)

Seed catalog: Carriers — *VENUSPRO Insurance* (VPRO) and *Summit Specialty* (SMT). Products — *Commercial Auto* (\$1,200 base), *General Liability* (\$850), *Workers Compensation* (\$3,200), *Business Owners / BOP* (\$1,450, Draft).

Step 1 — Create the form (Basic Info)

Open **Form Designer** → **Create Form**. This is the metadata that anchors versioning and lifecycle.

Field	Meaning	Values
Form Name *	Human title shown in the Library and runtime.	free text
Description	One-liner about the form's purpose.	free text
Carrier *	The insurer the form belongs to. Changing it re-points the Product.	from the Carrier catalog
Product *	The line of business; supplies base rate & coverages. The list is filtered to the chosen carrier.	from the Product catalog
Form Type *	The lifecycle role of the form.	Application · Supplemental · Quote · Endorsement · Cancellation · Claim · Renewal
Version *	Edition string (e.g. 1.1).	free text
Status *	Lifecycle state.	Draft · Published · Archived
Effective / Expiration Date	The window this form edition is valid for.	dates
Navigation Style	How respondents move through the form (also editable in <i>Workflow & Nav</i>).	Wizard · Tabs · Accordion · Single Page
Base Premium	The starting premium the rating engine multiplies/adds onto.	number (\$)

A "**Written for Carrier** → **Product**" banner and a live **metadata summary** (sections · fields · rules · validations · rating rules) keep you oriented. Header actions: **Publish** (flips status to Published) and **Save & Build Sections** (→ Section Builder).

Shortcut — draft the whole form with AI

The screenshot shows the VenusPro Form Platform interface. On the left is a dark sidebar with a menu. The main area is titled 'Form Designer > AI Generate'. It features a 'Describe your form' section with a text input containing an example prompt about a commercial auto application. Below this is a 'Generate Form' button and a list of example prompts. To the right, there are two informational boxes: 'HOW IT WORKS' and 'SETUP'.

VENUSPRO
FORM PLATFORM

Form Designer > AI Generate

Search forms...

Viewing as Admin Admin

FORM DESIGNER - AI

AI Form Generation

Describe a form in plain English and Claude drafts the sections, fields and types as editable metadata.

Describe your form
Claude Opus 4.8 · structured output

e.g. A commercial auto application for a trucking company, capturing business details, a schedule of vehicles, drivers, and coverage selections.

Generate Form or try an example --

Commercial auto application for a trucking... Homeowners insurance quote with property d... Auto claim — first notice of loss with inc... Workers compensation renewal questionnaire

HOW IT WORKS

1. Your description is sent to a small Express proxy (/api/generate-form).
2. It calls the Claude Messages API with a strict JSON schema (structured outputs).
3. The returned spec is normalized into editable form metadata.
4. Load it into the designer to refine sections, fields, logic and rating.

SETUP

Real AI needs the backend running with a key:

```
# .env
ANTHROPIC_API_KEY=sk-ant-...

npm run dev:full # web + AI Server
```

Without a key, generation falls back to a local heuristic draft so the flow still works.

Workspace: Dashboard, Form Library

Catalog: Carriers, Products, Guardrails

Form Designer: **AI Generate**, Create Form, Section Builder, Field Builder, Conditional Logic, Formula Builder, Validation Engine, Repeating Grid, Lookup Tables, Rule / Rating, Workflow & Nav

Runtime: Preview (Live), Review & Submit, Outputs

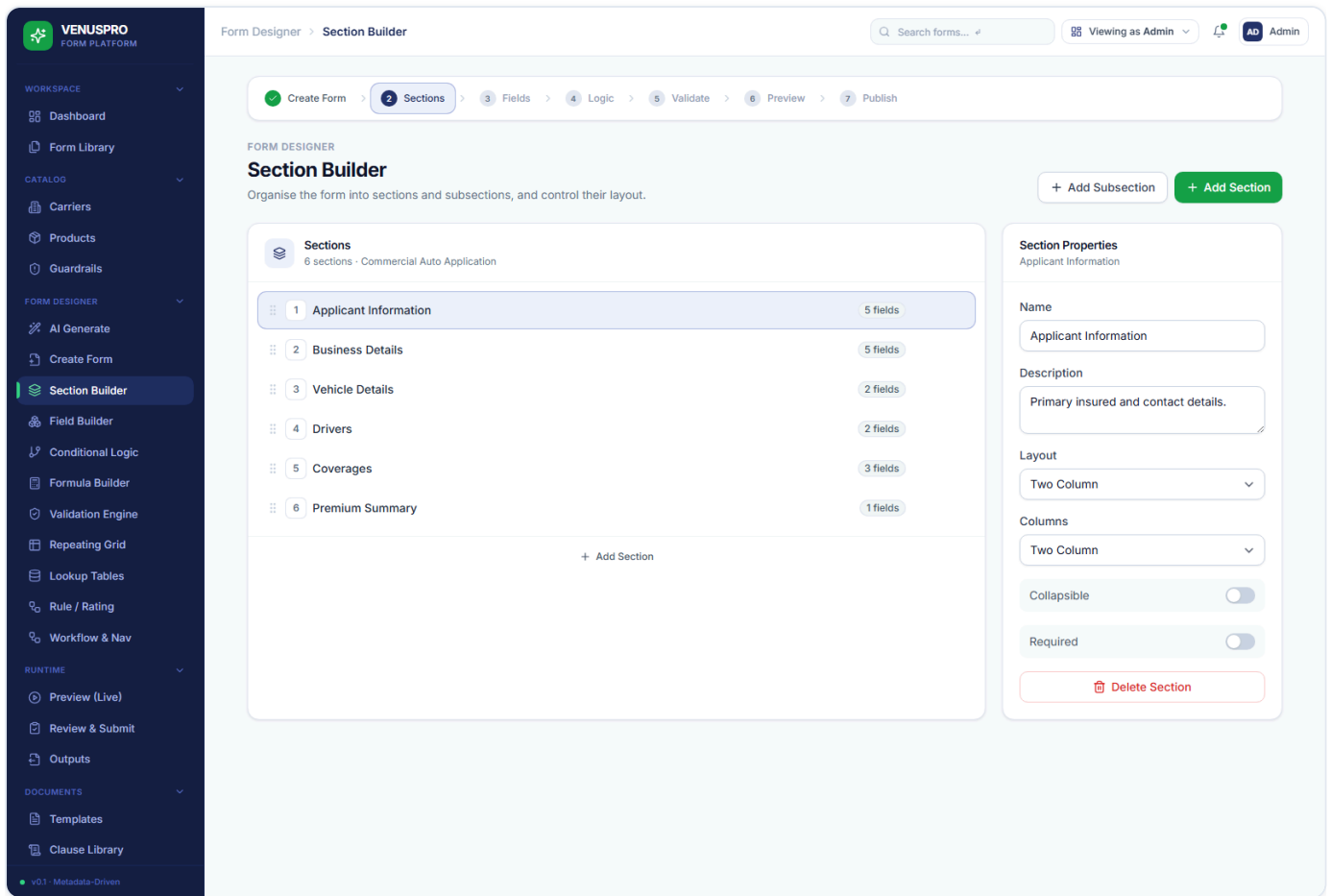
Documents: Templates, Clause Library

v0.1 · Metadata-Driven

AI Generate turns a plain-English description into editable form metadata. Type a prompt (or pick an example), **Generate Form**, review the proposed sections/fields, then **Load into Designer** to refine. It calls a small Express proxy (/api/generate-form) which uses the Claude Messages API with a strict JSON schema; **without an API key it falls back to a local heuristic draft** so the flow still works. See *Developer API* for setup.

Step 2 — Organise into Sections

Open **Form Designer** → **Section Builder**. Sections are the navigable chunks (wizard steps / tabs / accordion panels).



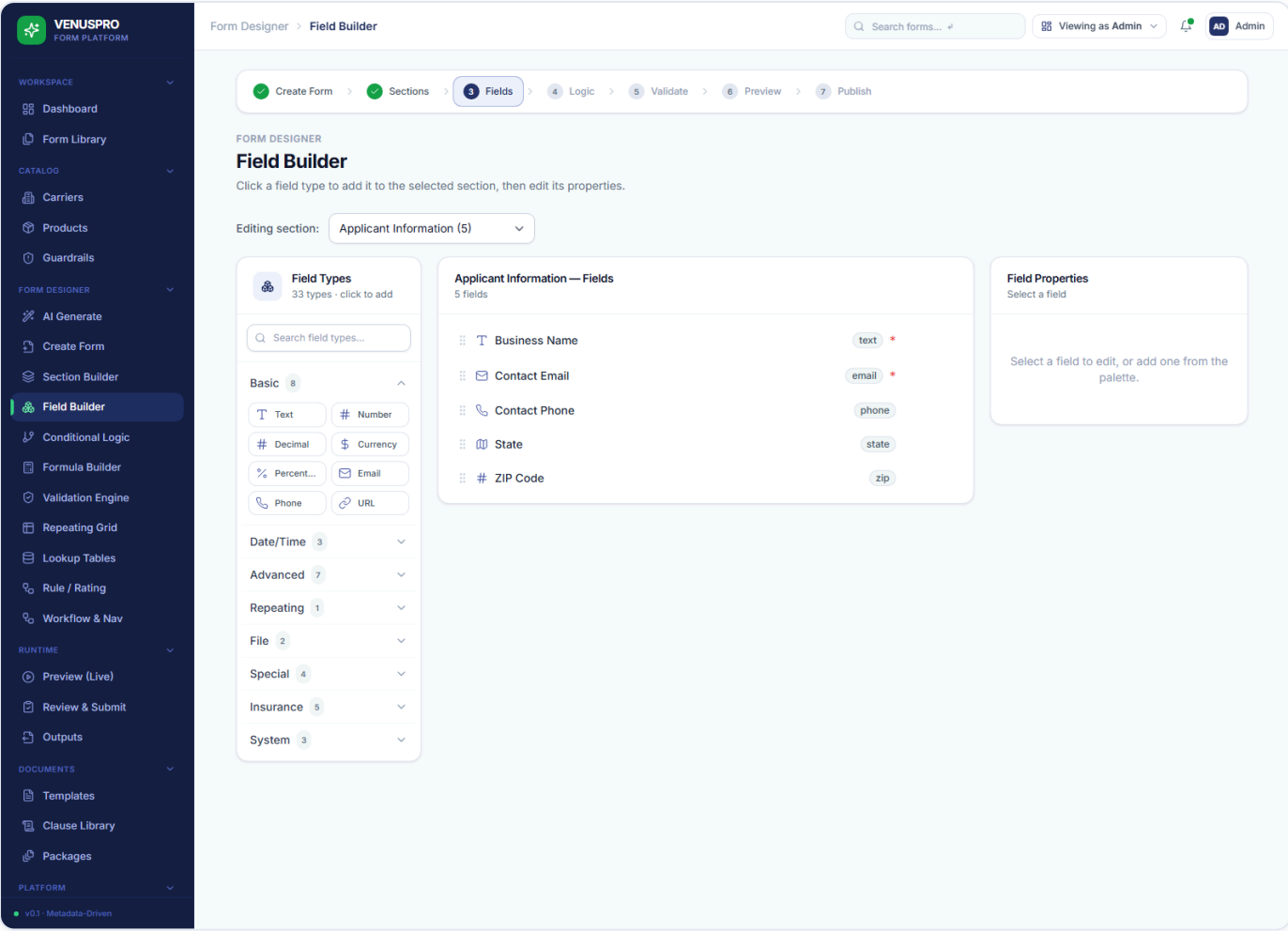
The **left tree** lists sections in order with a field count each; **drag the grip**, or use the ▲▼ buttons, to reorder; the trash icon deletes. Select a section to edit it on the right:

Property	Meaning	Values
Name	Section heading shown in the runtime.	free text
Description	Sub-heading / instructions.	free text
Layout	How fields lay out within the section.	Single Column · Two Column · Accordion
Columns	Grid width for the fields.	One Column · Two Column
Collapsible	Whether the section can collapse (accordion-style).	toggle
Required	Marks the section as mandatory (governance flag).	toggle

Add Section and **Add Subsection** are in the header. Subsections nest under a section for structure.

Step 3 — Add Fields

Open **Form Designer** → **Field Builder**. Pick the section to edit (top dropdown), click a type in the palette to append it, then edit its properties.



Left — the palette. 33 field types in 8 searchable, collapsible groups. Click a type to add it to the selected section.

Group	Types
Basic (8)	Text · Number · Decimal · Currency · Percentage · Email · Phone · URL
Date/Time (3)	Date · Date & Time · Time
Advanced (7)	Address · State · Country · Zip Code · Dropdown · Radio · Checkbox
Repeating (1)	Grid / Table
File (2)	File Upload · Image Upload
Special (4)	Signature · Rich Text · Label · Divider
Insurance (5)	Limit · Deductible · Coverage Selector · Class Code · Premium
System (3)	Hidden Field · System Field · Computed Field

Centre — fields in the section. Reorder by drag or ▲▼; delete with the trash icon; click a field to edit it.

Right — Field Properties (for the selected field):

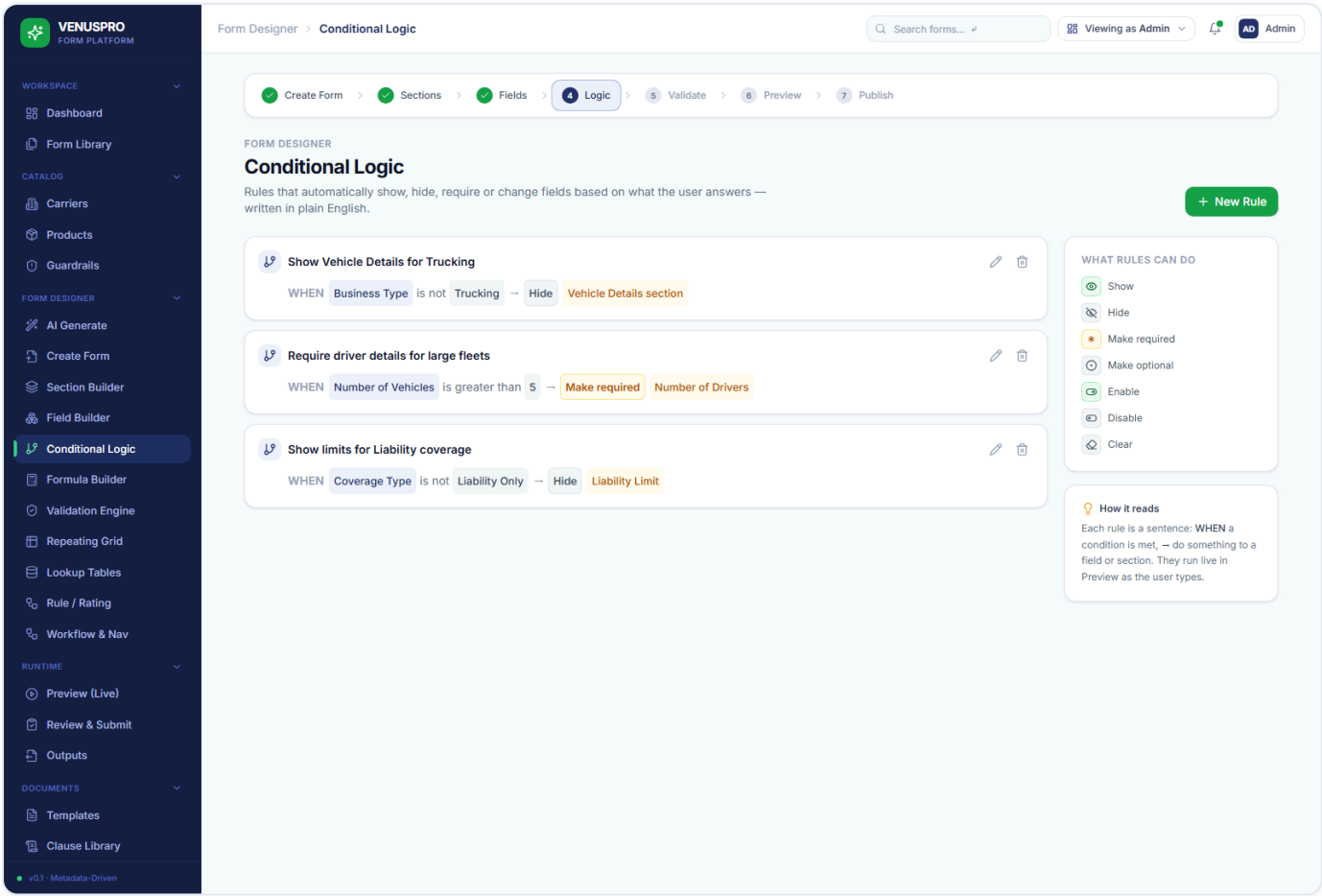
Property	Meaning	Applies to
Label	The question text shown to the user.	all
API Name	The machine key answers are stored under, and what formulas/conditions reference. (<i>spaces stripped</i>)	all
Help Text	A hint shown under the field.	all
Placeholder	Ghost text inside the input.	all
Column Span	1 or 2 — how wide the field is in a two-column section.	all
Prefix	Unit decoration, e.g. \$ or % .	all
Formula	The expression that computes the value (see Formula reference).	computed , premium
Options	The label→value choices (add/edit/remove rows).	select , radio , coverage , checkbox
Required	Field must be answered (gates submit as a blocker).	toggle
Read-only	Display-only (e.g. a computed premium).	toggle

Naming matters. Conditions, formulas, validations and rating rules all reference fields by **API Name**. Keep them stable and meaningful (`vehicleCount` , `annualRevenue`) — renaming an API name silently breaks any rule that points at the old one.

Step 4 — Logic

Three engines share the same condition model and run **live** in Preview.

4a. Conditional Logic — show / hide / require by answer



Each rule reads as a sentence: **WHEN** *conditions* → *action* on *targets*. Click **New Rule**:

Field	Meaning
Rule name	A label, e.g. "Show vehicles for trucking".
WHEN	The condition group (the shared condition editor — see Reference 4.2). Combine conditions with AND / OR / NOT .
THEN do this	The action: Show, Hide, Make required, Make optional, Enable, Disable, Clear .
to these fields / sections	The targets — check any sections and/or fields the action applies to.
Preview	A live plain-English read-back of the rule.

Seeded examples: *Show Vehicle Details for Trucking*, *Require driver details for large fleets* (`vehicleCount > 5`), *Show limits for Liability coverage*.

4b. Formula Builder — derive values live

The screenshot shows the VenusPro Form Platform interface, specifically the Formula Builder section. The left sidebar contains a navigation menu with categories: WORKSPACE (Dashboard, Form Library), CATALOG (Carriers, Products, Guardrails), FORM DESIGNER (AI Generate, Create Form, Section Builder, Field Builder, Conditional Logic, **Formula Builder**, Validation Engine, Repeating Grid, Lookup Tables, Rule / Rating, Workflow & Nav), RUNTIME (Preview (Live), Review & Submit, Outputs), and DOCUMENTS (Templates, Clause Library). The main area is titled 'Form Designer > Formula Builder' and includes a search bar and a user profile 'Admin'. A progress bar at the top shows steps: Create Form, Sections, Fields, **4. Logic**, Validate, Preview, and Publish. The 'FORM DESIGNER' section is titled 'Formula Engine' with the subtitle 'Derive values, premiums, tiers and eligibility — evaluated live against the current answers.' and a 'Load Sample Data' button. The 'Formula Editor' section shows an 'Expression' field with the formula `ROUND(1200 + vehicleCount * 150 + numEmployees * 25, 0)` and a 'Result' box displaying '1,200'. Below this is an 'EXAMPLE FORMULAS' section with several pre-defined formulas in input fields. On the right, there are two panels: 'FUNCTIONS' with a grid of function chips (SUM, AVG, MIN, MAX, ROUND, IF, COUNT, LOOKUP, CONCAT, TEXT, YEAR, MONTH, TODAY) and 'OUTPUTS' with chips for Field Values, Premium, Eligibility, Tier / Class, and Derived Data.

A live sandbox: type an **expression**, see the **result** computed against the current answers. References resolve by **API Name** (`vehicleCount` , `numEmployees`). Click a **Function** chip to insert it, or an **Example** to load it. Use these expressions in a field's **Formula** property (for `computed` / `premium` fields) or in `expression` -type validations.

Example: `ROUND(1200 + vehicleCount * 150 + numEmployees * 25, 0)` . Full function list in Reference 4.3.

4c. Rule / Rating Engine — price, qualify, route

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FORM PLATFORM

WORKSPACE

Dashboard

Form Library

CATALOG

Carriers

Products

Guardrails

FORM DESIGNER

AI Generate

Create Form

Section Builder

Field Builder

Conditional Logic

Formula Builder

Validation Engine

Repeating Grid

Lookup Tables

Rule / Rating

Workflow & Nav

RUNTIME

Preview (Live)

Review & Submit

Outputs

DOCUMENTS

Templates

Clause Library

v0.1

Metadata-Driven

Form Designer > Rule / Rating

Search forms...

Viewing as Admin

Admin

Create Form > Sections > Fields > **4 Logic** > 5 Validate > 6 Preview > 7 Publish

RULE & RATING ENGINE

Rule Builder

Shared IF-THEN engine — outputs premium, eligibility, tier, referrals and messages, evaluated live.

Load Sample Data

New Rule

Rating Rules

7 rules · live evaluation

RULE	IF (CONDITION)	THEN (OUTPUT)	LIVE
Trucking class surcharge	businessType = Trucking	Surcharge Trucking risk (×1.25)	
Large fleet surcharge	vehicleCount > 5	Surcharge Large fleet (×1.1)	
Full coverage uplift	coverageType = Full Coverage	Premium Full coverage (×1.4)	
Preferred tier for established revenue	annualRevenue > 10000000	Tier/Class Tier A	
New business referral	yearsInBusiness < 2	Referral New business — refer to underwriter	
Fleet too large for program	vehicleCount > 50	Eligibility Declined — fleet exceeds program limit	
Construction inspection notice	businessType = Construction	Message Construction class requires a site inspection before bind.	

LIVE RESULT

RATED PREMIUM

\$1,200

Tier: Standard Eligible

Base Premium

\$ 1200

OUTPUTS

Premium

Eligibility

Tier / Class

Referral

Messages

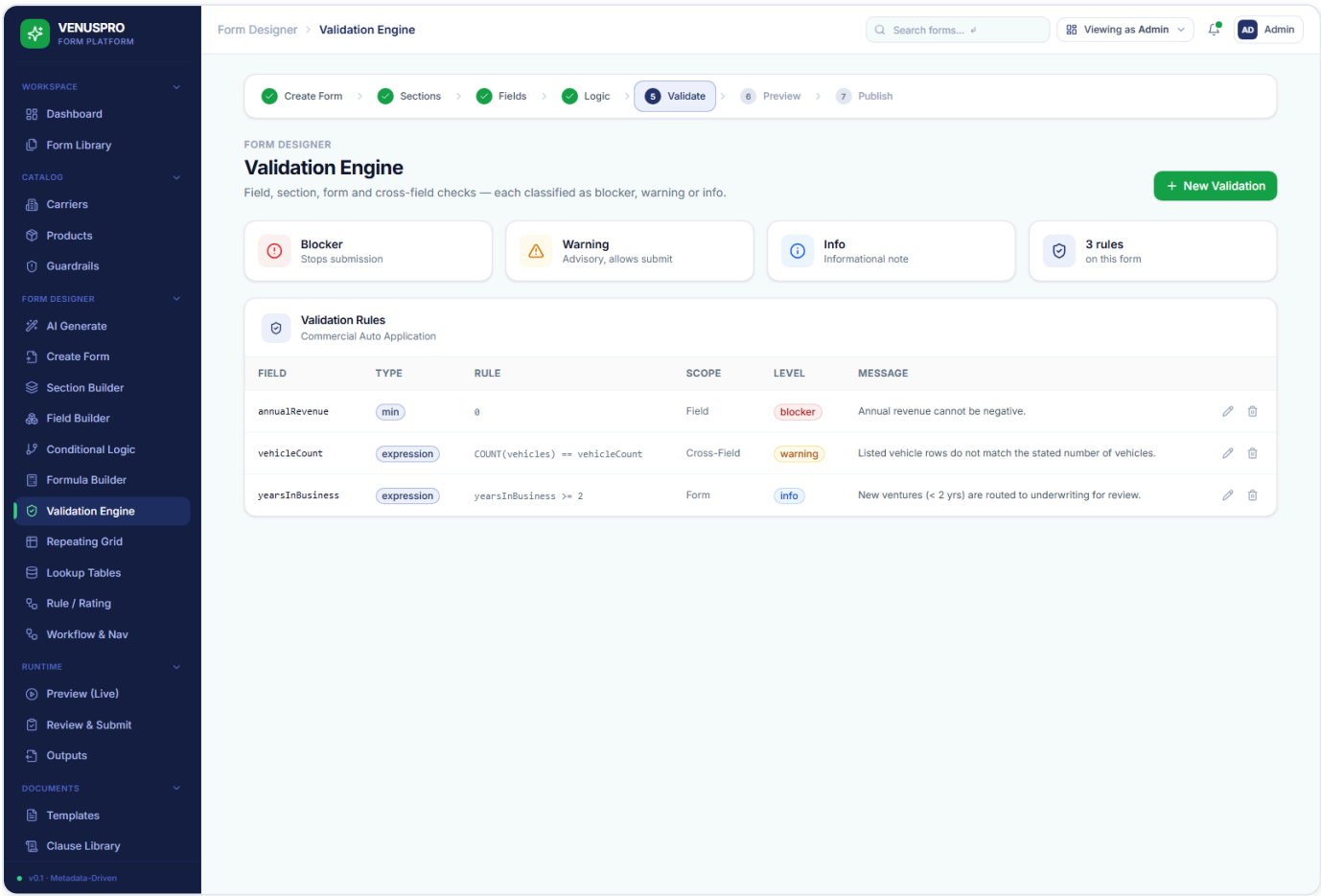
The shared IF-THEN engine that turns answers into a **rated result**. It starts from the form's **Base Premium** and applies each firing rule top-down; the right panel shows the **live result** (premium, tier, eligibility) and updates as you type. Click **New Rule**:

Field	Meaning	Values
Rule Name	A label, e.g. "Large fleet surcharge".	free text
IF — Conditions	The condition group (same editor as Conditional Logic).	—
Output	What the rule produces.	Premium · Surcharge · Tier/Class · Eligibility · Referral · Message
Result	The human-readable outcome, e.g. Tier B , Decline , a message.	free text
Effect (Premium/Surcharge only)	How it changes premium.	Multiplier (×) · Flat (+)
Amount (Premium/Surcharge only)	The multiplier or flat amount.	number

Seeded rules include a *Trucking class surcharge* (×1.25), *Large fleet surcharge* (×1.1), a *Preferred tier* for high revenue, a *new-business referral*, and a *fleet-too-large* decline. The **Base Premium** is editable here too. See Reference 4.4 for exactly how each output behaves.

Step 5 — Validate

5a. Validation Engine — per-form checks



Each validation is one check. Click **New Validation**:

Field	Meaning	Values
Field	The field (API name) the check applies to.	from the form's fields
Type	The kind of check.	required · min · max · minLength · maxLength · regex · expression
Value / Expression	The threshold, pattern, or (for <code>expression</code>) a formula that must be true to pass.	depends on type
Level	Severity. Blocker stops submit; Warning and Info are advisory.	Blocker · Warning · Info
Scope	Where it applies (labelling/grouping).	Field · Section · Form · Cross-Field
Message	The text shown when it fails.	free text

A field marked **Required** in the Field Builder already produces a blocker automatically — you only add a `required` *validation* for cross-field cases. `expression` validations are the cross-field workhorse, e.g. `COUNT(vehicles) == vehicleCount`. Only **blockers** gate submission; warnings/info let the user proceed.

5b. Guardrails — platform-wide governance

VENUSPRO

FORM PLATFORM

Catalog > Guardrails

Q Search forms...

Viewing as Admin

AD Admin

WORKSPACE

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PLATFORM

Global Guardrails

Platform-wide underwriting, rating and validation rules enforced on every quote & bind — on top of each form's own logic.

Reset to defaults

Underwriting eligibility

Appetite & territory

Enforce state availability

Block / decline a quote when the product isn't filed in the applicant's state.

Uses each product's filed states from the catalog.

Rating clamps

Premium floors & caps (priced forms)

Minimum premium

Floor every rated premium. 0 = off.

\$ 500

Maximum premium

Cap every rated premium. 0 = off.

\$ 0

Max discount

Premium can't fall below base – this %, 0 = off.

40 %

Referral & decline thresholds

Auto-route by size

Refer premium above

Premium over this → underwriter referral. 0 = off.

\$ 10000

Decline premium above

Premium over this → auto-decline (blocks bind). 0 = off.

\$ 50000

Refer vehicles above

Vehicle count over this → referral. 0 = off.

25

Global field validations

Applied to every form

Require signature to bind

A signature field must be completed before binding.

Effective date not in the past

Warn when an effective date is before today.

Expiration after effective

Block when expiration is on/before the effective date.

Validate email & phone

Check email/phone fields are well-formed across all forms.

v0.1 · Metadata-Driven

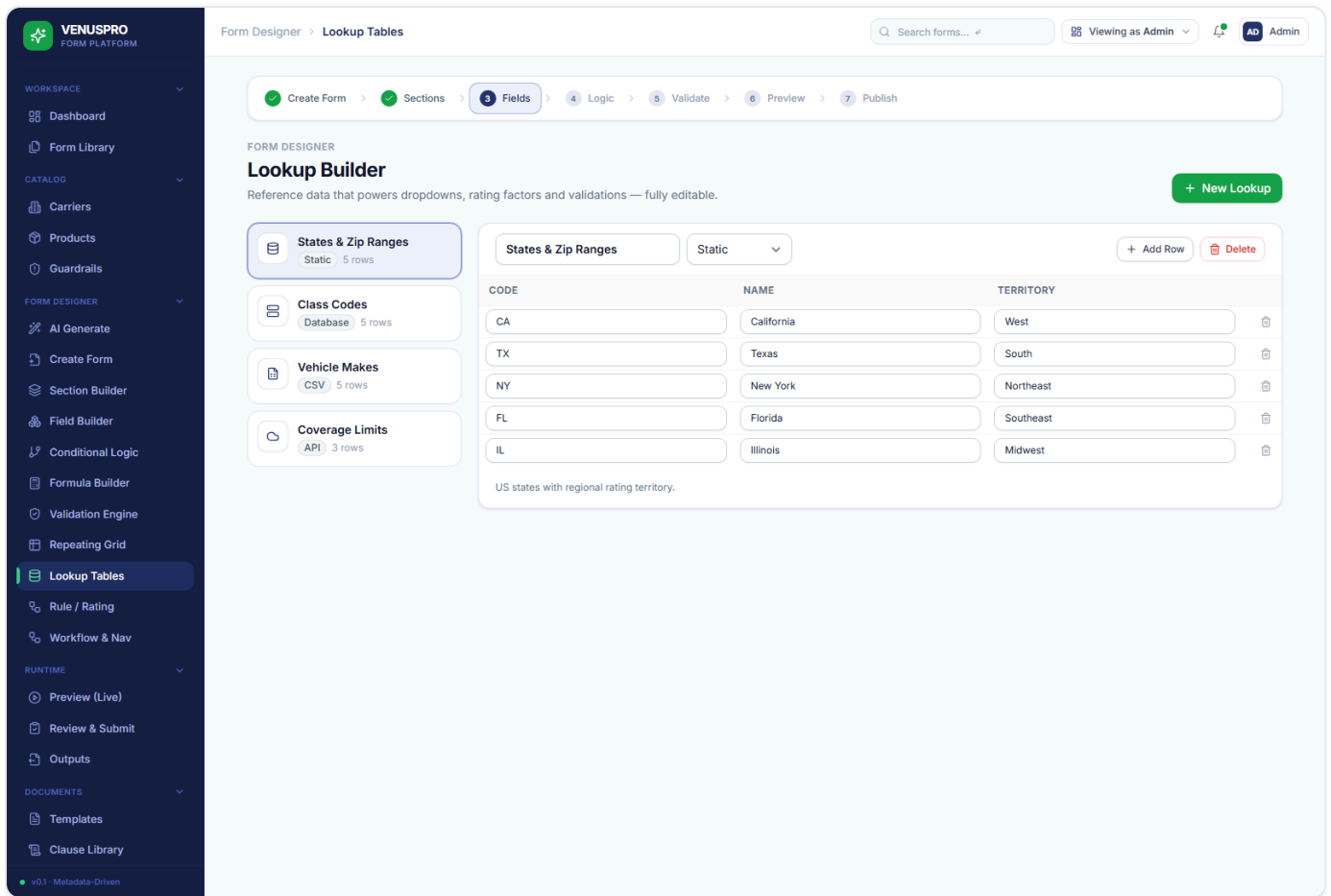
Guardrails (under *Catalog*) sit **on top of every form**, regardless of that form's own rules. They're configured once across four cards. Each finding is **block / refer / warn**, and the overall **decision** is Eligible → Referral → Declined (it can only escalate). A quote is **clear to bind** only when not Declined and no block finding exists.

Card	Setting	Default	Effect
Underwriting eligibility	Enforce state availability	on	Decline when the product isn't filed in the applicant's state.
Rating clamps (<i>priced forms</i>)	Minimum premium	\$500	Floor every rated premium (0 = off).
	Maximum premium	off	Cap every premium (0 = off).
	Max discount	40%	Premium can't fall below base – this %.
Referral & decline thresholds	Refer premium above	\$10,000	Over this → underwriter referral.
	Decline premium above	\$50,000	Over this → auto-decline (blocks bind).
	Refer vehicles above	25	Vehicle count over this → referral.
Global field validations	Require signature to bind	on	A signature field must be completed before bind.
	Effective date not in the past	on	Warn when effective date is before today.
	Expiration after effective	on	Block when expiration ≤ effective.
	Validate email & phone	on	Check email/phone fields are well-formed across all forms.

Reset to defaults restores the table above. Full evaluation order is in Reference 4.6.

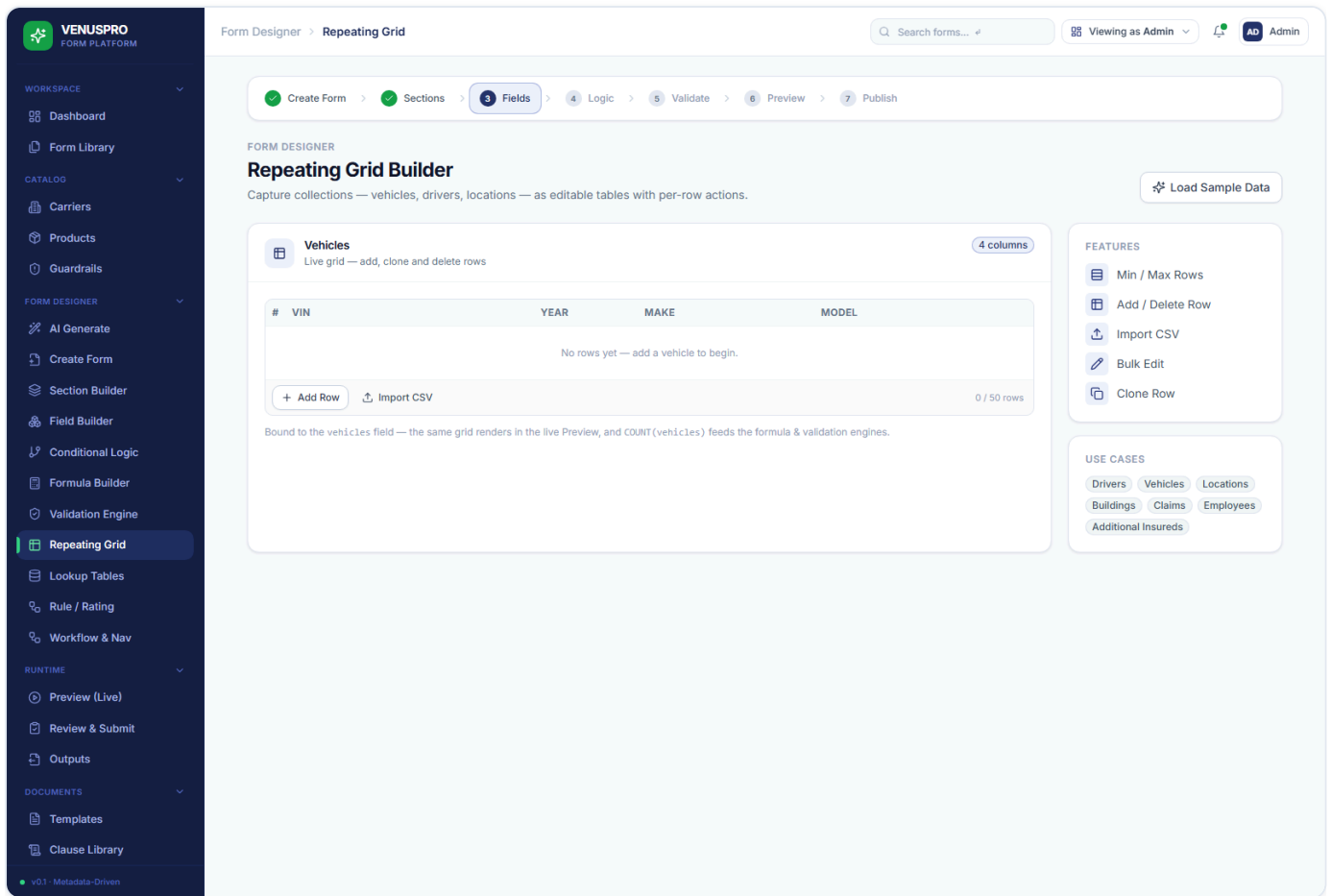
Supporting builders

Lookup Tables — editable reference data that powers dropdowns, rating factors and validations.



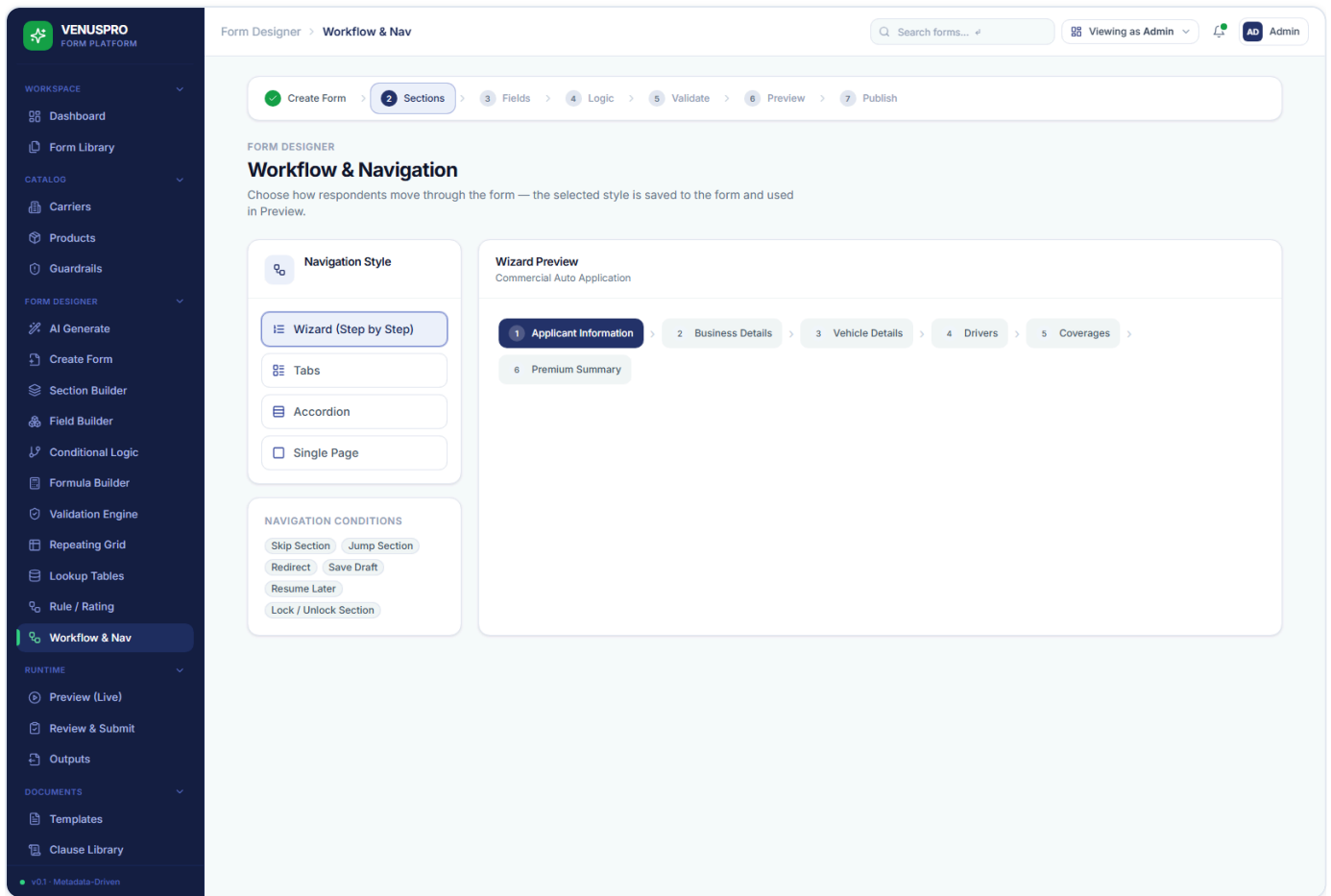
Pick a table on the left; edit its **Name**, **Source** (`Static` / `CSV` / `Database` / `API`), columns and rows on the right (cells are inline-editable; **Add Row** / delete per row). Seeded tables: *States & Zip Ranges*, *Class Codes*, *Vehicle Makes*, *Coverage Limits*. Reference a table from a formula with `LOOKUP(tableId, key, column)` .

Repeating Grid — capture collections (vehicles, drivers, locations) as editable tables.



This screen renders the first **Grid / Table** field in the form (add one from the Field Builder). Each grid has **columns** (with per-column type — text/number/select), **min/max rows**, and per-row **Add / Clone / Delete**. The row count feeds the engines via `COUNT(apiName)` — e.g. validate that listed vehicles match a stated count.

Workflow & Nav — choose how respondents move through the form.



Pick a **Navigation Style** — **Wizard (Step by Step)**, **Tabs**, **Accordion**, or **Single Page** — saved to the form and used in Preview; the right panel previews it. (The *Navigation Conditions* chips — Skip / Jump / Redirect / Save Draft / Resume Later / Lock-Unlock Section — are roadmap labels, not yet configurable.)

Step 6 — Preview against live data

VENUSPRO FORM PLATFORM

Runtime > Preview (Live)

Create Form > Sections > Fields > Logic > Validate > **Preview** > Publish

RUNTIME - LIVE

Form Preview

The form is rendered entirely from metadata — show/hide, calculations, validation and the navigation style below are all live.

Commercial Auto Application (v1.1) Published Wizard

Customer View Load Sample Data Reset

Desktop Tablet Mobile

1 Applicant Information 2 Business Details 3 Drivers 4 Coverages 5 Premium Summary

Applicant Information
Primary insured and contact details.

Business Name *
e.g. Summit Logistics LLC
Business Name is required.

Contact Email * **Contact Phone**
name@company.com (555) 123-4567
Contact Email is required.

State **ZIP Code**
Select... 90210

< Back Resolve 2 required fields to continue Next >

RATING ENGINE

RATED PREMIUM
\$1,200
base \$1,200
Tier: Standard Eligible

GUARDRAILS

DECISION Eligible
No guardrail issues.
Clear to bind

VALIDATION

Blockers 4
Warnings 0
Info 1
4 blockers to resolve

ACTIVE LOGIC

- Show Vehicle Details for Trucking
- Show limits for Liability coverage

Preview (Live) renders the form *entirely from metadata* in the chosen navigation style, with a **device** toggle (Desktop / Tablet / Mobile). **Load Sample Data** fills realistic answers; **Reset** clears them; **Customer View** opens the applicant-facing flow. The right **engine panel** is live:

- **Rating Engine** — rated premium (vs base), tier, eligibility, surcharges, referrals, messages.
- **Guardrails** — the decision, any premium clamp, findings, and the **Clear to bind / Bind blocked** status.
- **Validation** — blocker / warning / info counts and a *Ready to submit* pill.
- **Active Logic** — which conditional rules are currently firing.

This is where everything comes together: show/hide rules, formulas, the rating engine, validation and guardrails all run as you type. Notice in the screenshot that the seeded "Vehicle Details" section is hidden because the conditional rule keys off `businessType == Trucking` — that's branching working live.

Review & Submit

VENUSPRO

FORM PLATFORM

WORKSPACE

Dashboard

Form Library

CATALOG

Carriers

Products

Guardrails

FORM DESIGNER

AI Generate

Create Form

Section Builder

Field Builder

Conditional Logic

Formula Builder

Validation Engine

Repeating Grid

Lookup Tables

Rule / Rating

Workflow & Nav

RUNTIME

Preview (Live)

Review & Submit

Outputs

DOCUMENTS

Templates

Clause Library

Packages

PLATFORM

Versioning

v0.1 - Metadata-Driven

Runtime > Review & Submit

Search forms...

Viewing as Admin

Admin

Create Form > Sections > Fields > Logic > Validate > Preview > Publish

RUNTIME

Review & Submit

Confirm every section before submitting the application.

Back to Form Save Draft Generate PDF Submit

4 blocker(s) must be resolved before submitting. 0 warning(s), 1 info.

Applicant Information

Business Name

Contact Email

Contact Phone

State

ZIP Code

Edit

Business Details

Business Type

Years in Business

Annual Revenue

Number of Employees

Number of Vehicles

Edit

Drivers

Number of Drivers

Drivers

Edit

Coverages

Coverage Type

Deductible

Edit

Premium Summary

Estimated Base Premium

\$1,200

Edit

QUOTE SUMMARY

RATED PREMIUM

\$1,200

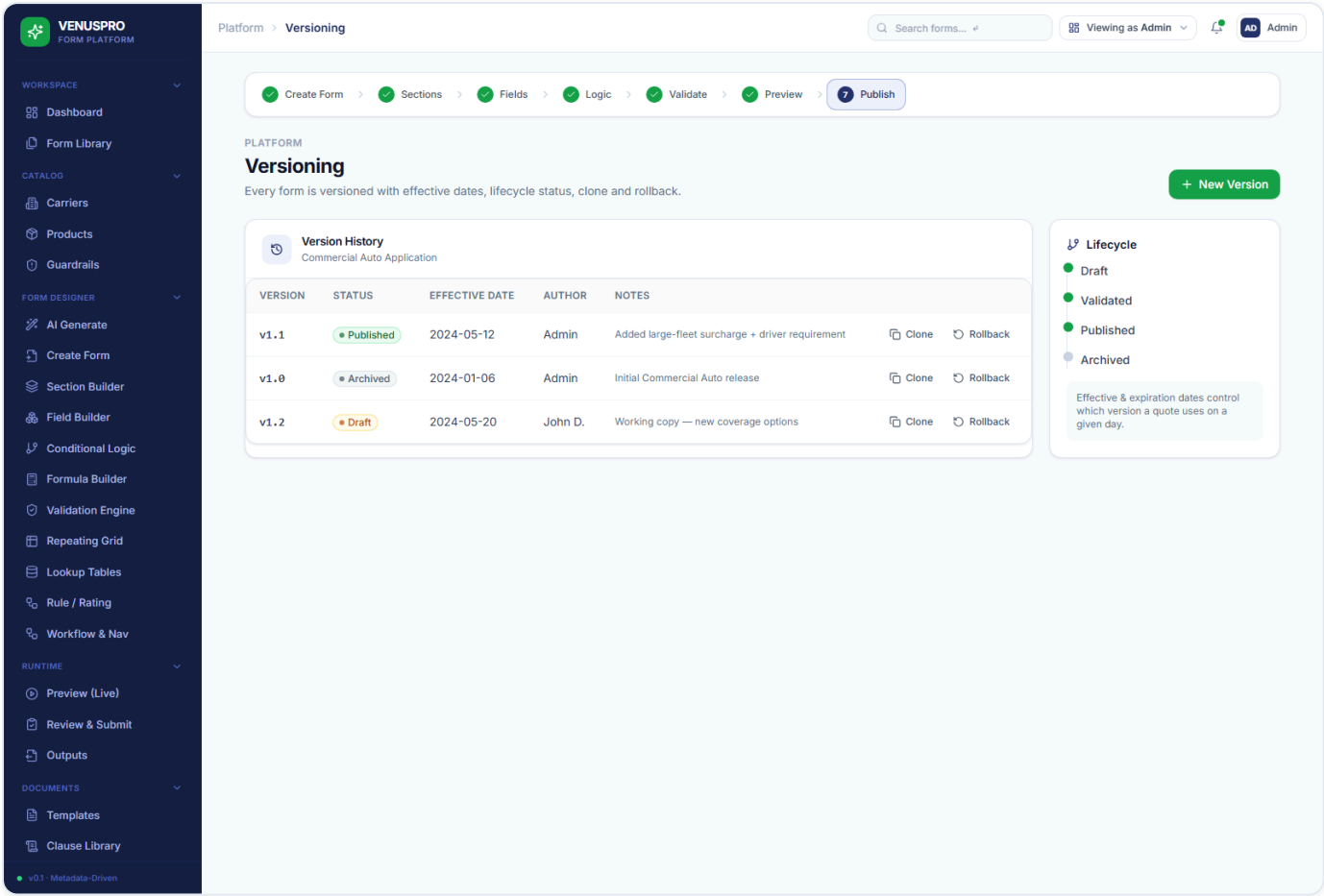
Tier: Standard Eligible

ADVISORIES

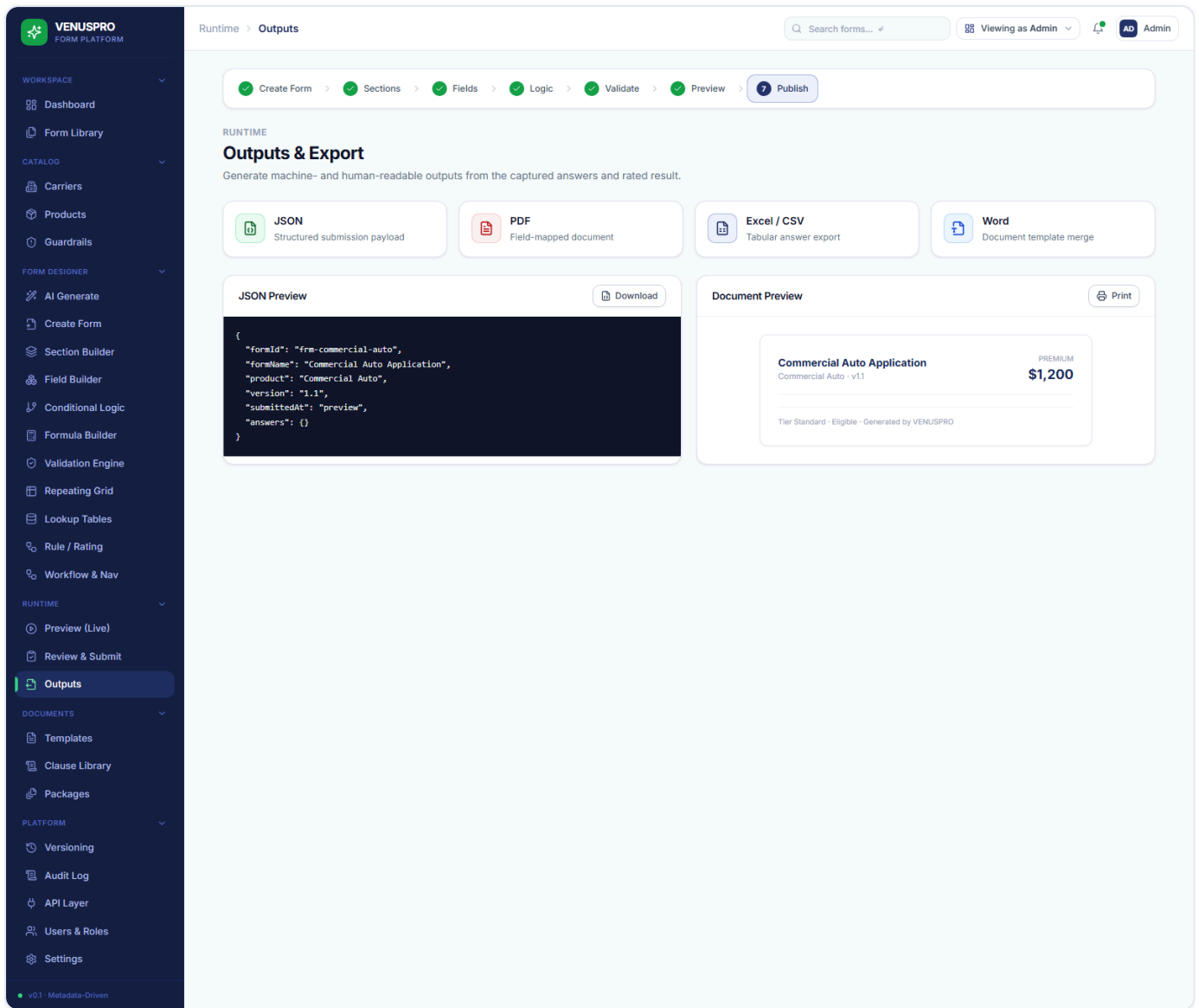
New ventures (< 2 yrs) are routed to underwriting for review.

A read-back of every visible section with a **Quote Summary** (premium, tier, eligibility) and **Advisories** (warnings/info). **Submit** is disabled until all blockers clear; submitting shows a confirmation modal with **Download JSON**. **Generate PDF** exports the field-mapped document.

Step 7 — Publish, Version & Export



Versioning lists the form's editions (Version, Status, Effective Date, Author, Notes, actions). **New Version / Clone** capture a full **snapshot** of the current form definition as a new Draft (marked *Restorable*); **Rollback** restores a snapshot and re-points the published edition. The lifecycle stepper reads **Draft** → **Validated** → **Published** → **Archived**; effective & expiration dates decide which version a quote uses on a given day.



Outputs generates machine- and human-readable artifacts from the captured answers + rated result: **JSON** (the submission payload), **PDF** (field-mapped document), **Excel / CSV** (tabular), and **Word** (print/merge), with live JSON and document previews.

4. Reference

4.1 Field-type catalog (33 types)

Group	Type · (label)
Basic	text Text · number Number · decimal Decimal · currency Currency · percentage Percentage · email Email · phone Phone · url URL
Date/Time	date Date · datetime Date & Time · time Time
Advanced	address Address · state State · country Country · zip Zip Code · select Dropdown · radio Radio · checkbox Checkbox
Repeating	grid Grid / Table
File	file File Upload · image Image Upload
Special	signature Signature · richtext Rich Text · label Label · divider Divider
Insurance	limit Limit · deductible Deductible · coverage Coverage Selector · classcode Class Code · premium Premium
System	hidden Hidden Field · system System Field · computed Computed Field

Money types (`currency` , `limit` , `deductible` , `premium`) default a `$` prefix; `percentage` defaults `%` ; `computed` / `premium` carry a **Formula**; `select` / `radio` / `coverage` / `checkbox` carry **Options**.

4.2 Condition operators (the WHEN / IF editor)

Used identically by Conditional Logic and the Rule/Rating engine. A condition is `field OPERATOR value` ; group them with **AND** (all), **OR** (any) or **NOT** (none), and nest groups.

Operator	Reads as	Compares
<code>equals</code>	is	string equality
<code>notEquals</code>	is not	string inequality
<code>greaterThan</code>	is greater than	numeric
<code>lessThan</code>	is less than	numeric
<code>greaterOrEqual</code>	is at least	numeric
<code>lessOrEqual</code>	is at most	numeric
<code>contains</code>	contains	case-insensitive substring
<code>isEmpty</code>	is empty	no value / empty array
<code>isNotEmpty</code>	is not empty	has a value

An **empty** condition group evaluates **true** (the rule always fires). A reference to a grid field yields its **row count**.

4.3 Formula functions & syntax

A safe expression evaluator (no `eval`). Bare identifiers resolve against live answers by **API Name** (missing $\rightarrow$ `0` ; arrays $\rightarrow$ `length`). Operators: `+` `-` `*` `/` , comparisons `>` `<` `>=` `<=` `==` `!=` , parentheses, string literals, `TRUE` / `FALSE` . `+` concatenates when either side is non-numeric.

Function	Does
<code>SUM(...)</code> / <code>AVG(...)</code> / <code>MIN(...)</code> / <code>MAX(...)</code>	Aggregate numeric args.
<code>ROUND(x, d?)</code>	Round <code>x</code> to <code>d</code> decimals (default 0).
<code>IF(cond, a, b)</code>	<code>a</code> if <code>cond</code> truthy, else <code>b</code> .
<code>COUNT(ref)</code>	Array length, or count of non-empty args.
<code>LOOKUP(tableId, key, column)</code>	First row whose any value equals <code>key</code> $\rightarrow$ its <code>column</code> .
<code>CONCAT(...)</code>	Join args as text.
<code>TEXT(x)</code>	Stringify.
<code>YEAR(x)</code> / <code>MONTH(x)</code>	Pull year/month from an ISO date string.
<code>TODAY()</code>	Today (<i>currently a stub — returns empty</i>).

Examples: `ROUND(1200 + vehicleCount * 150 + numEmployees * 25, 0) * annualRevenue * 0.0008` .
`IF(vehicleCount > 5, 1.1, 1.0) * COUNT(vehicles)` .

4.4 Rating outputs & effects

`rate()` starts at the form's **Base Premium** and applies each firing rule top-down; the final premium is rounded.

Output	Behaviour
Premium / Surcharge	<code>multiplier</code> $\rightarrow$ <code>premium *= amount</code> ; <code>flat</code> $\rightarrow$ <code>premium += amount</code> . Listed as a surcharge.
Tier/Class	Sets the result tier (last firing rule wins).
Eligibility	If the result text matches <i>declin...</i> $\rightarrow$ Declined ; <i>refer...</i> $\rightarrow$ Referral .
Referral	Adds a referral note; promotes eligibility to Referral if still Eligible.
Message	Adds an advisory message.

Result shape: `{ basePremium, premium, tier (default "Standard"), eligibility (Eligible/Referral/Declined), surcharges[], referrals[], messages[] }` .

4.5 Validation types & levels

Type	Fails when
<code>required</code>	value is empty
<code>min</code> / <code>max</code>	numeric value below / above the threshold
<code>minLength</code> / <code>maxLength</code>	text length below / above the threshold
<code>regex</code>	value doesn't match the pattern
<code>expression</code>	the formula doesn't evaluate truthy (cross-field)

Levels: **blocker** (stops submit) · **warning** · **info** (both advisory). Required fields auto-emit a blocker; only blockers gate `canSubmit`.

4.6 Guardrail evaluation order

1. **State availability** — product not filed in the answered `state` → **block** (Declined).
2. **Rating clamps** (priced forms): max-discount floor → minimum-premium floor → maximum-premium cap (each a **warn** when it adjusts).
3. **Referral / decline thresholds** vs the adjusted premium → **block** (decline) or **refer**; vehicle-count threshold → **refer**.
4. **Email / phone** well-formedness (email fail = **block**, phone fail = **warn**).
5. **Dates** — effective-in-past = **warn**; expiration ≤ effective = **block**.
6. **Signature** required-to-bind and missing → **block**.
7. **canBind** = decision ≠ Declined *and* no `block` finding.

4.7 Statuses & navigation styles

Form status	Meaning
Draft	Editable; not live.
Published	Live edition.
Archived	Retired; kept for history.

Navigation styles: `Wizard` (stepper, one section at a time, gated *Next*), `Tabs` (free tab switching, per-tab blocker counts), `Accordion` (collapsible panels), `Single Page` (everything open).

5. Worked example — a Commercial Auto Application

Goal: a Commercial Auto application that captures the business, a schedule of vehicles & drivers, and coverages — prices itself, shows trucking-only sections, and won't bind oversized fleets.

1. **Catalog** — in **Products**, confirm *Commercial Auto* (\$1,200 base, states CA/TX/NY/FL/IL, Liability/Physical Damage/Medical/UM coverages).

2. **Create Form** — Name "*Commercial Auto Application*"; Carrier *VENUSPRO Insurance*; Product *Commercial Auto*; Type **Application**; Navigation **Wizard**; Base Premium **1200**.
3. **Sections** — Applicant Information · Business Details · Vehicle Details · Drivers · Coverages · Premium Summary (6 sections).
4. **Fields** — add `businessType` (Dropdown incl. *Trucking*), `vehicleCount` (Number), a `vehicles` **Grid**, `annualRevenue` (Currency), a `coverageType` (Coverage Selector), `liabilityLimit` (Limit), and a read-only **Premium** field with formula `ROUND(1200 + vehicleCount * 150 + numEmployees * 25, 0)`.
5. **Logic** —
 - *Conditional*: `WHEN businessType is not "Trucking" → Hide the Vehicle Details section`; `WHEN vehicleCount is greater than 5 → Make required the driver fields`.
 - *Rating*: `businessType == Trucking → Surcharge x1.25`; `vehicleCount > 5 → Surcharge x1.1`; `vehicleCount > 50 → Eligibility "Declined – fleet exceeds program limit"`.
6. **Validate** — `annualRevenue min 0` (blocker); cross-field `expression: COUNT(vehicles) == vehicleCount` (warning).
7. **Guardrails** — leave state-availability **on** (so non-CA/TX/NY/FL/IL declines), minimum premium \$500, decline above \$50,000.
8. **Preview** — *Load Sample Data*; confirm the premium rates live, the Vehicle Details section appears only for Trucking, and the engine panel reports **Clear to bind**.
9. **Publish & Version** — set Status **Published**; in **Versioning**, *Clone* to snapshot v1.1.

The seed form ships exactly this way: open **Form Library** → **Commercial Auto Application** → **Preview**. Because `businessType` starts empty ($\neq$ *Trucking*), the **Vehicle Details** section is hidden — the wizard shows 5 of 6 sections until you pick *Trucking*.

6. Developer API

The **API Layer** page documents the form contract (mocked on the bundled Express server):



This page isn't working

localhost is currently unable to handle this request.

HTTP ERROR 500

Reload

GET /form/{id} · POST /form/{id}/save (draft answers) · POST /form/{id}/submit · POST /form/{id}/validate · POST /form/{id}/calculate (premium) · GET /rules · POST /files/upload · POST /documents/pdf · GET /lookups/{table} · GET /form/{id}/versions .

The live backend (`npm run server` , port 8787; Vite proxies `/api`) currently implements **AI generation**:

- `POST /api/generate-form { prompt }` → a structured form spec via the Claude Messages API (model `claude-opus-4-8` , JSON-schema structured output).
- `GET /api/health` → `{ ok, ai }` .

```
# .env
ANTHROPIC_API_KEY=sk-ant-...

npm run dev:full # web + AI server
```

Without a key, **AI Generate** falls back to a local heuristic draft.

7. What's not yet configurable (gaps & roadmap)

Honest list, roughly by priority.

P1 — multi-form & lifecycle

1. **One active form at a time.** The Designer edits a single in-memory form; loading another from the Library (or AI-generating) **replaces** it. There's no side-by-side editing or per-form isolation — snapshot via *Versioning* → *Clone* before switching.
2. **No payment step.** The runtime is **Preview** → **Review** → **Submit** (a confirmation modal) and **Publish** is a *status* action — there is **no checkout/billing/payment** screen anywhere. (Listed because it's commonly expected alongside "preview / publish".)
3. **setValue action isn't in the UI.** The conditional-logic engine supports a *Set value* action, but the rule editor only exposes Show/Hide/Require/Optional/Enable/Disable/Clear.

P2 — builder depth 4. **Subsections are structural only** — you can add them, but the Field Builder edits fields at the section level (no dedicated subsection field palette). 5. **Workflow "Navigation Conditions"** (Skip / Jump / Redirect / Save Draft / Resume Later / Lock-Unlock) are display-only labels, not yet wired. 6. **Grid "Import CSV"** currently just adds a blank row (stub); `TODAY()` in formulas returns empty (stub).

P3 — catalog & data 7. **Carriers & Products are read-only** in the UI (seeded catalog; no add/edit screen). 8. **Lookup sources** (`CSV` / `Database` / `API`) are labels — rows are edited inline; there's no live fetch/sync. 9. **Guardrails key off conventional field names** (`state` , `vehicleCount` , and email/phone/effective/expiration by type or name pattern) rather than an explicit mapping per form.

P4 — production-readiness 10. **Persistence is browser localStorage** (demo) — production needs server storage and a real submit/rate/validate API behind the documented contract. 11. **Rating is a single top-down pass** — no rating tables beyond lookups, no taxes/fees/installments. 12. **Roles aren't enforced** — the persona switcher and Users & Roles page are presentational.

8. Cheat sheet

I want to...	Go to...
Start a new form / set product & type	Create Form
Draft a form from a description	AI Generate → <i>Load into Designer</i>
Organise the form into steps	Section Builder
Add or edit a question	Field Builder → pick a type → edit properties
Show/hide/require a field by answer	Conditional Logic
Compute a value live	Formula Builder (or a field's <i>Formula</i> property)
Price the form / set eligibility & tiers	Rule / Rating
Add a data check	Validation Engine
Enforce a rule across every form	Guardrails
Manage reference data (states, class codes...)	Lookup Tables
Capture a collection (vehicles, drivers)	Field Builder → add a <i>Grid</i> field → Repeating Grid
Choose Wizard / Tabs / Accordion / Single Page	Workflow & Nav (or <i>Create Form</i> → <i>Navigation Style</i>)
Test it live (with the engines running)	Preview (Live)
Review everything before submit	Review & Submit
Export JSON / PDF / CSV / Word	Outputs
Snapshot or roll back the form	Versioning
See which carrier/product a form is for	Carriers / Products

Generated for VENUSPRO — Form Builder.